B. Strange Machine

time limit per test 1.5 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1000 | Tags: binary search, brute force, greedy, implementation | https://codeforces.com/problemset/problem/2156/B

You are given n machines arranged in a circle, where n is at most 20. Each machine is either of type A or type B. The machines are numbered clockwise from 1 to n, and the type of the i-th machine is denoted by s_i. Each machine takes an integer x and updates it according to its type:

- Type A: Decrease x by 1. Formally, update $x := x - 1$.
- Type B: Replace x with the floor of half its value. Formally, update $x := \lfloor x/2 \rfloor$, where $\lfloor y \rfloor$ denotes the floor of y, which is the greatest integer less than or equal to y.

You are given q queries, each consisting of a single integer a. In each query, you start at machine 1 holding an integer a. Each second, the following two actions occur in order:

1. The current machine updates a according to its type.
2. Then, move one step clockwise to the next machine. Formally

- If you are at machine i where $1 \le i \le n - 1$, move to machine $i + 1$.
- If you are at machine n, move to machine 1.

This process continues until your integer a becomes 0. For each query, determine the number of seconds required for a to reach 0.

Note that all queries are independent of each other.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains two integers n and q ($1 \le n \le 20$, $1 \le q \le 10^4$) — the number of machines, and the number of queries, respectively.

The second line of each test case contains a string s ($|s| = n$ and $s_i = \mathtt{A}$ or $\mathtt{B}$) — the types of the machines.

The third line of each test case contains q integers $a_1, a_2, \ldots, a_q$ ($1 \le a_i \le 10^9$) — the initial integer of each query.

Note that there are no constraints on the sum of n over all test cases.

It is guaranteed that the sum of q over all test cases does not exceed 10^4.

Output

For each test case, output q integers representing the answers to each query.

Example

Input

```
3
2 2
BA
3 4
1 1
B
20
6 4
BAABBA
2 8 32 95
```

Output

```
2
3
5
2
5
8
11
```

Note

In the first test case, the queries are as follows:

- Query 1: $a = 3$ Therefore, it takes 2 seconds for a to reach 0.

1. Start at machine 1. Machine 1 replaces a with the floor of half its value. Now, $a = \lfloor 3/2 \rfloor = 1$.
2. Move to machine 2. Machine 2 decreases a by 1. Now, $a = 1 - 1 = 0$.

- Query 2: $a = 4$ Therefore, it takes 3 seconds for a to reach 0.

1. Start at machine 1. Machine 1 replaces a with the floor of half its value. Now, $a = \lfloor 4/2 \rfloor = 2$.
2. Move to machine 2. Machine 2 decreases a by 1. Now, $a = 2 - 1 = 1$.
3. Move back to machine 1. Machine 1 replaces a with the floor of half its value. Now, $a = \lfloor 1/2 \rfloor = 0$.

In the second test case, there is only one query with $a = 20$:

1. Start at machine 1. Machine 1 replaces a with the floor of half its value. Now, $a = \lfloor (20)/2 \rfloor = 10$.
2. Move to machine 1. Machine 1 replaces a with the floor of half its value. Now, $a = \lfloor (10)/2 \rfloor = 5$.
3. Move to machine 1. Machine 1 replaces a with the floor of half its value. Now, $a = \lfloor 5/2 \rfloor = 2$.
4. Move to machine 1. Machine 1 replaces a with the floor of half its value. Now, $a = \lfloor 2/2 \rfloor = 1$.
5. Move to machine 1. Machine 1 replaces a with the floor of half its value. Now, $a = \lfloor 1/2 \rfloor = 0$.

Therefore, it takes 5 seconds for a to reach 0.